Лабораторная работа 8

Оптимизация

Цвелев С.А. НПИбд-01-22

Цель работы

Основная цель работы — освоить пакеты Julia для решения задач оптимизации.

Решение задачи линейного программирования

```
[3]: model = Model(GLPK.Optimizer)

@variable(model, 0 <= x1 <= 10)
@variable(model,x2 >= 0)
@variable(model, x3 >= 0)

@constraint(model, -x1 + x2 + x3 <= -5)
@constraint(model, x1 + 3x2 - 7x3 <= 10)

@objective(model, Max, x1 + 2x2 + 5x3)

optimize!(model)

println("x1 = ", value(x1), " ", "x2 = ", value(x2), " ", "x3 = ", value(x3))
```

```
x1 = 10.0 x2 = 0.0 x3 = 5.0
```

Линейное программирование. Использование массивов

```
[5]: A = [-1 1 3; 1 3 -7]
     b = [-5; 10]
     c = [1; 2; 5]

     model = Model(GLPK.Optimizer)

     @variable(model, x[1:3] >= 0)
     @constraint(model, x[1] <= 10)
     @constraint(model, A * x .<=b)
     @objective(model, Max, c' * x)

     optimize!(model)

     println("x = ", value.(x))

     x = [10.0, 2.1875, 0.9375]
```

Выпуклое программирование

```
[6]: Random.seed!(0)

m, n = 30, 10
A = randn(m, n)
b = randn(m)

x = Convex.Variable(n)

problem = minimize(sumsquares(A * x - b), [x >= 0])
solve!(problem, SCS.Optimizer)

println("status = ", problem.status)
println("x = ", vec(x.value))
```

```
------------------------------------------------------------------
               SCS v3.2.11 - Splitting Conic Solver
        (c) Brendan O'Donoghue, Stanford University, 2012
------------------------------------------------------------------
problem:  variables n: 11, constraints m: 42
cones:    l: linear vars: 10
          q: soc vars: 32, qsize: 1
settings: eps_abs: 1.0e-04, eps_rel: 1.0e-04, eps_infeas: 1.0e-07
          alpha: 1.50, scale: 1.00e-01, adaptive_scale: 1
          max_iters: 100000, normalize: 1, rho_x: 1.00e-06
          acceleration_lookback: 10, acceleration_interval: 10
          compiled with openmp parallelization enabled
lin-sys:  sparse-direct-amd-qdldl
          nnz(A): 312, nnz(P): 0
------------------------------------------------------------------
 iter | pri res | dua res |   gap   |   obj   |  scale  | time (s)
------------------------------------------------------------------
     0| 6.19e+01  1.00e+00  8.76e+01 -4.28e+01  1.00e-01  2.07e-03
   250| 4.58e-04  4.81e-05  5.89e-04  2.47e+01  4.30e-01  6.43e-03
------------------------------------------------------------------
status:  solved
timings: total: 6.43e-03s = setup: 1.43e-04s + solve: 6.29e-03s
         lin-sys: 2.30e-04s, cones: 3.89e-05s, accel: 1.44e-03s
------------------------------------------------------------------
objective = 24.721028
------------------------------------------------------------------
status =
[ Info: [Convex.jl] Compilation finished: 4.64 seconds, 663.955 MiB of memory allocated
OPTIMAL
x = [0.00012365892959841494, -1.5997035443830413e-5, 0.000458021537585501, 0.23013380134682224, 0.14518273315878613, 2.0390039799587006e-5, 0.19591480979384623, 0.009438689011214816, 0.2643872879158339, 0.00023663855011374737]
```

План приготовления кофе

```
[7]: model = Model(GLPK.Optimizer)

@variable(model, raf >= 0, Int)
@variable(model, cap >= 0, Int)

@constraint(model, 40raf + 30cap <= 500)
@constraint(model, 140raf + 120cap <= 2000)
@constraint(model, 5raf == 40)

@objective(model, Max, 400raf + 300cap)

optimize!(model)

println("Раф = ", value(raf), " ", "Капучино = ", value(cap))
```

```
Раф = 8.0 Капучино = 6.0
```